

D&G Pricing Analysis

Evaluating Customer-Level Optimised Pricing for New Business

Pricing Analyst Case Study • Executive Summary Report

KEY RECOMMENDATION

Adopt the **@22% customer-level optimised pricing strategy** in place of the current ASIS approach. It generates substantially more revenue per offer with no statistically proven cost to conversion, though it should be rolled out alongside a proper live test given the caveats detailed in this report.

Prepared by Siraj Tausif Shaik

dg-pricing-analysis-case-study • github.com/SiriJ079

OVERVIEW

Executive Summary

Domestic & General currently prices new appliance-protection-plan offers using a fixed, unadjusted rate ('ASIS'). This project tests whether switching to a customer-level optimised pricing model, offered at two different strategy settings, @22% and @23%, would improve the business outcome, using a sample of 8,865 contact-centre offers. The analysis looks at conversion rate, revenue, how sensitive customers are to price (elasticity), whether the pricing models are well-calibrated, and whether a customer's claims history affects their likelihood to buy.

@22%
Recommended strategy

£6.42
Avg. revenue uplift per sale
(@22%)

2.4×
Conversion boost from a prior
claim

p = 0.61
Conversion difference is not
significant

Headline findings

- **@22% is the clear revenue winner.** It produced £6.42 of extra revenue per sale on average, and £5,545 in total across the sample, more than double @23%'s £2.92 and £2,535.
- **That revenue gain does not come at a proven cost to conversion.** A chi-square test found no statistically significant difference in conversion rate between the three strategies ($p = 0.61$), so ASIS's slightly higher raw conversion number (23.3% vs 22.0%) cannot be relied on as a genuine strategy effect.
- **Customers do respond to price, but the direction flips depending on how you slice the data.** Comparing each strategy against ASIS, conversion clearly falls as price rises, exactly as basic economic theory predicts. But comparing low- versus high-priced customers *within* the same strategy shows the opposite pattern, a sign that the pricing model itself is quietly influencing who gets offered a higher price.
- **Claims history is a powerful, independent signal.** Customers with a previous claim convert at roughly 2.4 times the rate of those without one (48.1% vs 20.0%), and this holds up even after accounting for whether they already own a plan.
- **The underlying data has some quality issues worth flagging,** most notably that ASIS's 'predicted conversion rate' field is a placeholder value of 1.0 rather than a genuine model output, which limits how fairly all three strategies can be compared on model accuracy.

METHODOLOGY

How the Data Was Prepared

The raw dataset covers contact-centre offers for appliance protection plans, priced under three strategies: **ASIS FEE** (the existing fixed-rate approach) and two customer-level optimised strategies, **@22%** and **@23%**, which adjust the price up or down for each customer based on a predictive model. Before any analysis, the data was cleaned in a dedicated notebook (`01_dataset_cleaning.ipynb`) to make sure comparisons between strategies would be fair and not distorted by formatting issues or bad records.

Cleaning steps applied

- Standardised column names to lowercase and converted date fields (offer date, purchase date) and numeric fields (premiums, price difference, claims, plan counts) to their correct data types, since inconsistent types would silently break later calculations.
- Removed exact duplicate rows to avoid double-counting any offer.
- Dropped rows missing the fields essential for comparing strategies (pricing point, sale outcome, base rate, offered premium, price difference), while deliberately keeping rows with missing claims data, since claims history is only needed for one specific part of the analysis and removing those rows everywhere would have thrown away useful data unnecessarily.
- Added two data-quality flags rather than silently deleting anything: one for non-positive prices, and one checking that offered premium minus base rate roughly equals the recorded price difference. No rows tripped the invalid-price flag in the final dataset, which is a reassuring sign the pricing fields are internally consistent.
- Saved the cleaned dataset separately from the raw file, so every later notebook works from the same trusted version.

Result: 8,865 of the original 8,867 rows were retained, meaning almost no data was lost at the cleaning stage, a good sign that the raw extract was already reasonably well-formed.

SECTION A

Exploratory Data Analysis

Before comparing the three pricing strategies head-to-head, it is important to check that the groups being compared are actually similar in every other respect, otherwise any difference in outcome could just reflect who was offered which strategy, not the strategy itself.

- **Offer volumes:** 1,115 ASIS offers, 3,923 @22% offers and 3,827 @23% offers were sampled. The two optimised strategies are represented in similar numbers, giving a reasonably balanced comparison between them.
- **Customer and appliance profiles were broadly comparable** across all three strategies: average purchase price (£398-£411), item age, and plan-ownership rates (around 20% in every group) did not differ meaningfully, which supports treating any outcome difference as a genuine effect of the pricing strategy rather than a difference in who was offered it.
- **price_diff is exactly zero for every ASIS row**, which makes sense: ASIS applies no customer-level adjustment by design. @22% and @23% both show a spread of price increases and decreases, visible as the two boxes in Figure 1 sitting away from the dashed zero-line that marks ASIS.
- **A data quality issue was found in the predicted-conversion field:** for every ASIS row, 'predicted conversion rate' is a constant value of 1.0, a placeholder rather than a genuine model output. This means

ASIS cannot be fairly included in any comparison of how well-calibrated each strategy's pricing model is (see Section D).

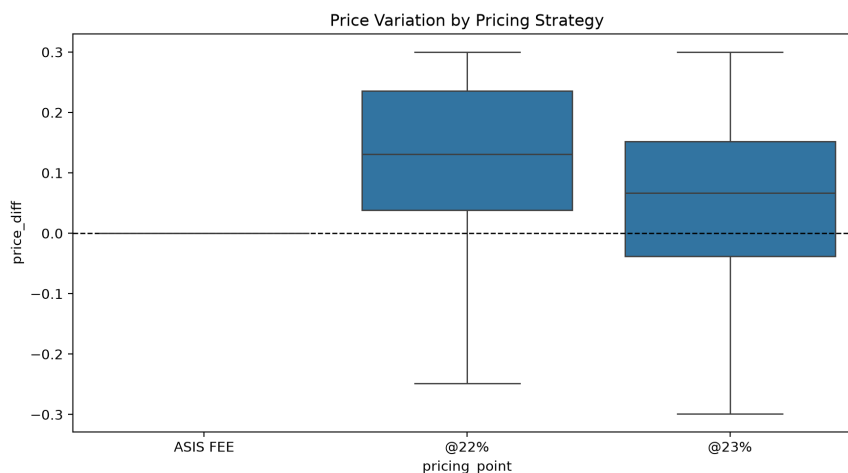


Figure 1: Spread of price adjustment (price_diff) applied under each strategy, relative to the ASIS baseline of zero.

The correlation matrix below (Figure 2) was used to sanity-check relationships between all the key numeric fields before running any formal statistical tests. A few things stand out: base rate and offered premium are very strongly related (0.91), as expected since the offered premium is mostly built from the base rate; plan-related fields (plan count, active plans, cancelled plans) cluster together strongly (0.45-0.85), suggesting they largely capture the same underlying customer behaviour; and claims count and claim amount are highly correlated (0.79), which is intuitive, more claims tend to mean a higher total claim amount.

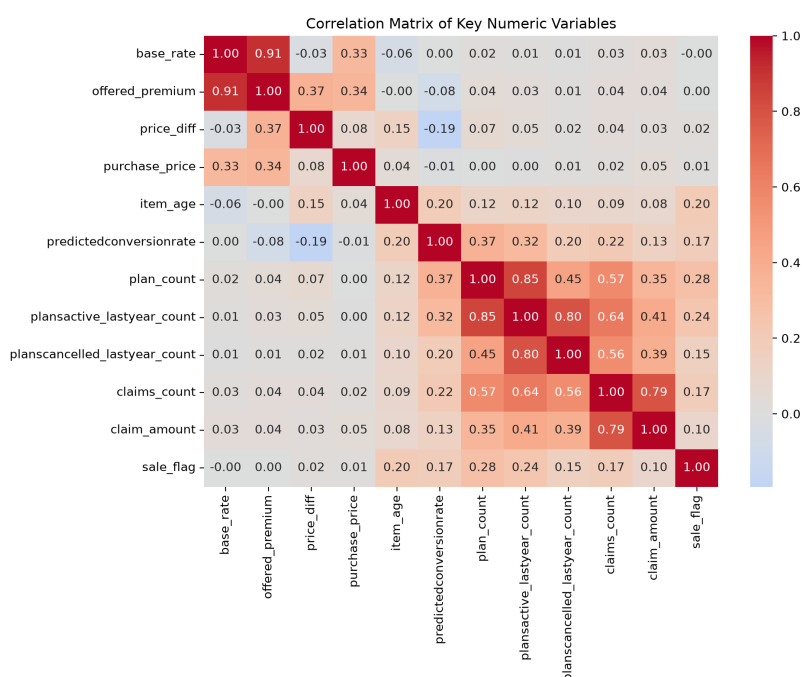


Figure 2: Correlation matrix across all key numeric variables in the cleaned dataset.

SECTION B

Pricing Strategy Performance

The core comparison of the three strategies is summarised below. ASIS is the current baseline; @22% and @23% are the two customer-level optimised alternatives being evaluated as replacements.

Strategy	Offers	Avg Base Rate	Avg Offered Prem.	Avg Sold Prem.	Avg Price Incr.	Conversion
ASIS FEE	1,115	£48.97	£48.97	£49.57	£0.000	23.3%
@22%	3,923	£49.20	£55.21	£55.75	£0.124	22.0%
@23%	3,827	£49.01	£51.98	£51.48	£0.060	22.7%

Read on its own, ASIS looks like it converts best (23.3% vs 22.0% and 22.7%). But raw percentages like this can be misleading without a statistical test to check whether the gap is real or just noise from sampling.

Statistical testing

- **Conversion (a yes/no outcome):** a chi-square test of independence across all three strategies returned $p = 0.61$, far above the usual 0.05 significance threshold. In plain terms, there is no strong statistical evidence that conversion rate genuinely differs by strategy, the small gaps seen in the table above are consistent with random variation.
- **Offered premium and price increase (continuous numbers):** Welch's t-tests were run between every pair of strategies. Every single comparison came back highly significant (all $p < 0.001$), confirming the strategies genuinely do price very differently from one another. This matters because it means the strategies are meaningfully distinct pricing treatments, even though that pricing difference is not currently translating into a proven conversion difference.

WHY THIS MATTERS

Because conversion is statistically flat across strategies but revenue is not (see Section D), the business case for switching strategy has to rest on the revenue evidence rather than a conversion improvement. This is an important nuance carried through to the final recommendation.

SECTION C

Price Elasticity

Elasticity measures how sensitive conversion is to a change in price, essentially, how much conversion moves for a given percentage change in premium. Two versions of this were calculated: a **between-group** comparison (each optimised strategy versus the ASIS baseline) and a **within-group** comparison (customers offered a low price versus a high price, inside the same strategy).

Comparison	Elasticity
@22% vs ASIS	-0.436
@23% vs ASIS	-0.464
@22% (low vs high price)	+0.660
@23% (low vs high price)	+0.171

- **Between-group elasticity is negative for both strategies, exactly as economic theory predicts:** @22% is -0.44 and @23% is -0.46 versus ASIS. A higher price relative to the ASIS baseline is associated with lower conversion, a sensible, expected result.
- **Within-group elasticity flips to positive for both strategies** (@22%: +0.66, @23%: +0.17), the opposite direction. On the surface this looks like customers convert *more* when charged *more*, which does not make economic sense taken at face value.
- **The likely explanation for @22% is a selection effect, not a real price response.** A weak positive correlation ($r = +0.11$) was found between the price adjustment and the model's own predicted conversion rate: the optimisation model tends to assign slightly higher prices to customers it already believes are more likely to buy. This means the model is partly pricing based on its own prediction of who will convert, so the positive within-group elasticity is reflecting that bias rather than customers genuinely preferring higher prices.
- **For @23%, this explanation does not fully hold** (correlation of only -0.04 with predicted conversion). None of the other variables tested, plan status, claims, item age, or plan count, fully explained the positive result either, so this remains a partially open question worth investigating further with additional data.

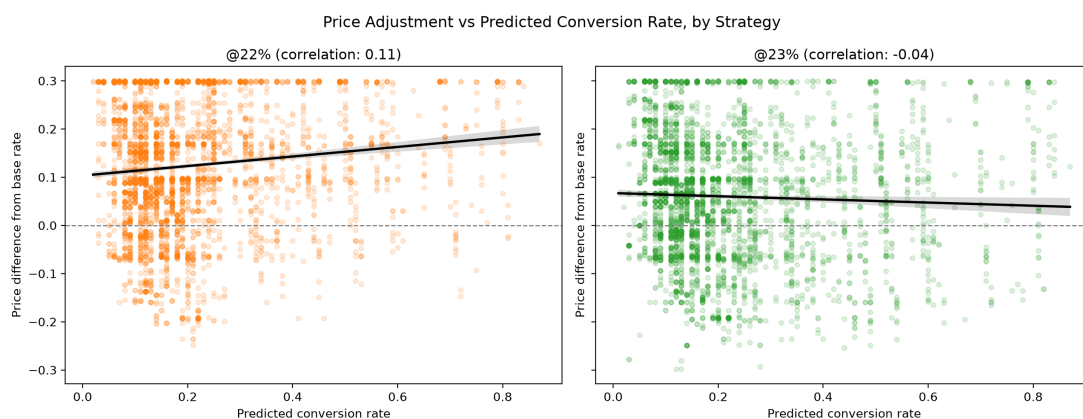


Figure 3: Price adjustment plotted against the model's predicted conversion rate, by strategy. The upward-sloping trend line for @22% (left) illustrates the selection effect described above.

SECTION D

Strategy Ranking and Model Bias

Pulling the statistical and revenue evidence together, this section ranks the three strategies across several different criteria, since 'which strategy is best' depends on which outcome matters most to the business.

Criterion	Best Strategy	Value
Statistical: conversion difference vs ASIS	@23%	0.227
Statistical: premium difference vs ASIS	@22%	55.21
Revenue: avg realised uplift per sale	@22%	6.418
Revenue: total realised uplift	@22%	5,545.08

Is @22% the winner across the board?

- **Revenue:** @22% wins decisively on both average uplift per sale (£6.42 vs £2.92) and total realised uplift across the sample (£5,545 vs £2,535). This is the strongest and clearest piece of evidence in @22%'s favour.
- **Premium:** @22% also produces the highest average offered premium of the two optimised strategies, consistent with it taking a bolder pricing approach.
- **Conversion:** @23% edges out @22% numerically (22.7% vs 22.0%), but as shown in Section B, this difference is not statistically significant overall ($p = 0.61$), so it should not be read as a genuine advantage for @23%.

Model calibration and bias

Strategy	Avg. Predicted Conversion	Actual Conversion	Calibration Gap
@22%	0.212	0.220	+0.0079
@23%	0.223	0.227	+0.0040

- Both optimised models are well calibrated: the gap between what each model predicted and what actually happened is small for @22% (0.008) and even smaller for @23% (0.004). ASIS is excluded here since its predicted-conversion field is the placeholder value flagged in Section A.
- However, calibration on average is not the same as being unbiased in *how* prices are set. The elasticity work in Section C shows @22%'s pricing decisions are mildly linked to its own conversion predictions, a selection bias that a simple average calibration check would not catch on its own.

CAVEAT TO FLAG

@22%'s revenue advantage is genuine, but its pricing logic shows a mild selection bias that should be validated with a proper randomised or holdout test before scaling the strategy significantly, rather than relying solely on this observational sample.

SECTION E

Claims History and Conversion

One question outside the direct strategy comparison was whether a customer's claims history relates to how likely they are to buy a new plan. This turned out to be one of the strongest patterns in the whole dataset.

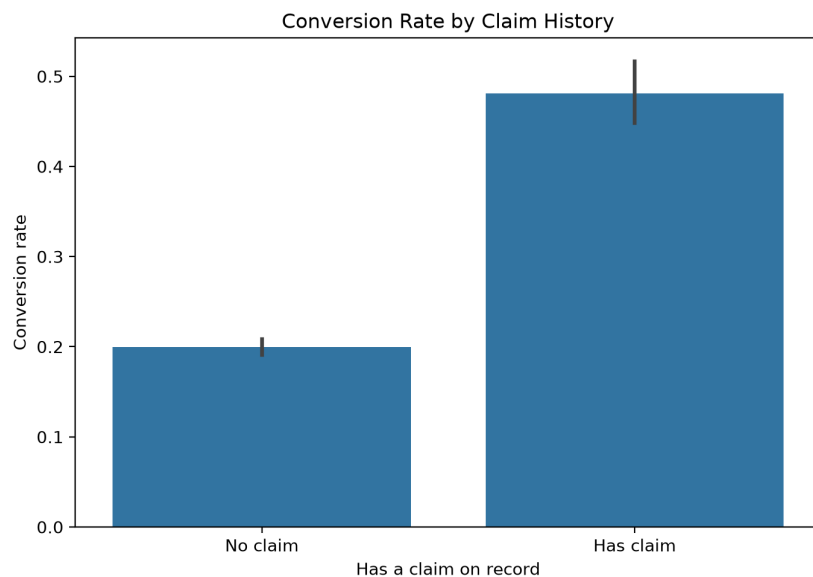
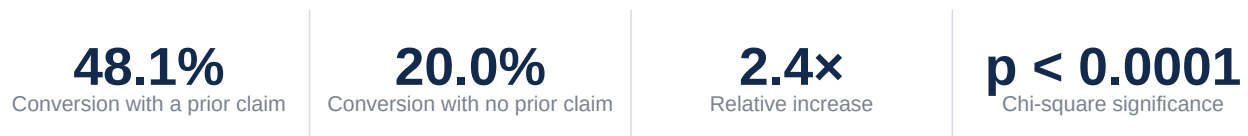


Figure 4: Conversion rate by claim history, with 95% confidence intervals. The non-overlapping intervals confirm this is a real, not a random, difference.

- A chi-square test confirmed this relationship is highly statistically significant ($p < 0.0001$), and the gap held up consistently across all three pricing strategies (ASIS: 50.5% vs 20.4%; @22%: 48.4% vs 19.5%; @23%: 47.0% vs 20.3%), ruling out the idea that claim holders were simply being steered toward whichever strategy converts best.
- Claim holders are disproportionately likely to already hold a plan (84.8% of them do, versus 13.4% of non-claimants), which makes intuitive sense, you generally need an active plan to make a claim in the first place. This raised the question of whether claims are simply a proxy for plan ownership rather than an independent effect.
- Controlling for plan ownership confirmed claims still matter on their own: among customers with *no* existing plan, having a claim on record still roughly doubles conversion (16.3% to 33.3%). Among customers who already hold a plan, claims still add a smaller but real boost (43.9% to 50.7%).
- The most likely explanation is behavioural rather than purely statistical: a customer who has recently experienced the value of their cover through a successful claim is probably more receptive to buying further protection, since the benefit feels concrete rather than abstract.

OPPORTUNITY FOR THE BUSINESS

Claims history is a genuinely new, independent segmentation variable that is not yet built into the pricing models. Incorporating it, particularly for non-plan customers where the effect is largest, is a promising and low-risk next step for improving targeting.

SECTION F

Final Recommendation

Bringing together the evidence from every section above, the recommendation is to move away from the current ASIS pricing approach and adopt the @22% customer-level optimised strategy, while continuing to monitor @23% as a lower-risk fallback.

RECOMMENDATION

Adopt @22% as the primary pricing strategy, replacing ASIS. The revenue case is strong and the conversion risk is not statistically proven, but this should be implemented alongside a proper live test rather than a full, immediate rollout, given the caveats below.

Why @22%

- It generates substantially more revenue than @23%: £6.42 average uplift per sale versus £2.92, and £5,545 total uplift versus £2,535 across the sample.
- The chi-square test found no statistically significant difference in conversion rate between strategies ($p = 0.61$), meaning this revenue gain does not come at a meaningful, provable cost to conversion volume.
- Both optimised strategies showed reasonable calibration between predicted and actual conversion (gaps of 0.008 and 0.004 respectively), suggesting the underlying models are not grossly miscalibrated on average.

Caveats to flag to the business

- @22%'s pricing decisions are weakly but measurably linked to its own predicted conversion rate ($r = +0.11$), meaning higher-priced customers within that strategy were already somewhat more likely to convert. This mild selection bias should be investigated further, ideally with a proper randomised or holdout test, before scaling @22% significantly.
- Conversion differences between strategies are not statistically significant at the overall level ($p = 0.61$), so this recommendation rests more heavily on the revenue evidence than on a proven conversion advantage.
- Claims history is a strong, independent driver of conversion (48.1% vs 20.0%, $p < 0.0001$), robust even after controlling for plan ownership. This is a genuinely new and actionable segmentation variable that is not yet used in the pricing models.
- ASIS's predicted-conversion field was found to be a constant placeholder value (1.0) rather than a genuine model output, meaning ASIS could not be included in the model-calibration comparison. This should be corrected in future data extracts to allow a fairer three-way comparison.

Suggested next steps

- Run a properly powered A/B test (or extend a current live test) comparing @22% against @23%, specifically designed to detect a conversion difference of the size observed here (around 2 percentage points), since the current sample was not statistically conclusive.
- Investigate incorporating claims history and plan status jointly into the pricing model's customer segmentation, given both showed large and partly independent conversion effects.
- Fix the ASIS predicted-conversion data gap so future analyses can properly assess model bias for all three strategies on an equal footing.

This analysis was completed end-to-end in Python (pandas, seaborn, scipy) as part of the Domestic & General Pricing Analyst case study. Full code, notebooks and outputs are available in the accompanying GitHub repository.